

University of Malakand



8th Spring-Semester Examination 2025 Fundamentals of Data Mining (BS Computer Science)

Time Allowed: 2:30hrs

Max. Marks: 60

Note: Questions 1 is compulsory and attempt any Four questions from remaining five questions.

Q1. Write short note/Definition of the following:

(20)

- a) Define Data Mining.
- b) Define Outlier Detection.
- c) What is Genetic Algorithm?
- d) Discuss any one data mining tool.

$21 \rightarrow \frac{21 - \text{min}}{\text{max} - \text{min}}$

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Q2. Describe the main concepts of data mining and data pre-processing.

(10)

Q3. Explain missing data handling, normalization, and discretization methods.

(10)

Q4. Discuss classification, clustering and association rule mining techniques.

(10)

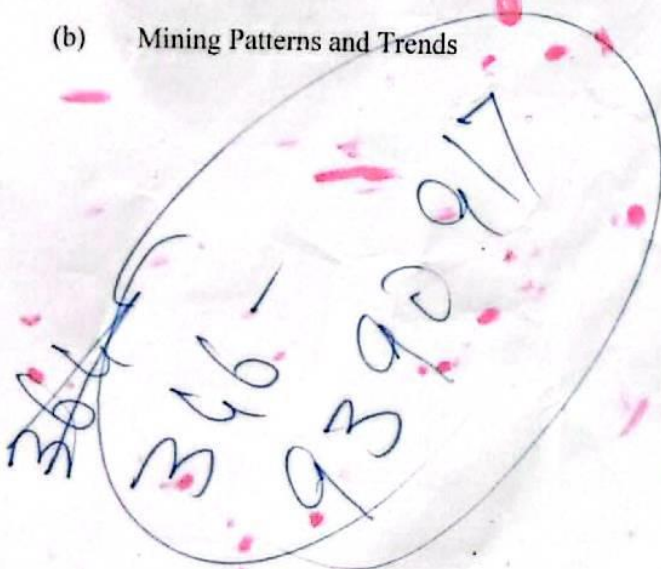
Q5. Explain the role of neural networks, fuzzy logic and genetic algorithms in data mining.

(10)

Q6. Write notes on the following:

(10)

- (a) Decision Trees
- (b) Mining Patterns and Trends



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8th Spring Semester Examination 2025

Human Resource Management (BS Computer Science)

Time Allowed: 2:30hrs

Max. Marks: 60

Note: Questions 1 is compulsory and attempt any Four questions from remaining five questions.

Q1. Write short note/Definition of the following: (20)

- a) Importance of Human Resource Management
- b) Define recruitment
- c) What is meant by initial screening of applicants?
- d) What is meant by job interview?

Q2. Please discuss external and organization environment? (10)

Q3. Please write detail note on HR planning? (10)

Q4. What is meant by employees performance evaluation? Also discuss key steps in employee's performance evaluation? (10)

Q5. What is job analysis? Discuss key steps in job analysis? (10)

Q6. As an HR manager what types of benefits and services will you recommend for employees wellbeing? (10)

$$n = \frac{a+b}{h} = \frac{0+1}{0.5} = 2$$

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8th Spring Semester Examination 2025
Numerical Computing (BS Computer Science)

Max. Marks: 60

Time Allowed: 2:30hrs

Note: Questions 1 is compulsory and attempt any Four questions from remaining five questions.

Q1. Write short note/Definition of the following:

(20)

- Discuss the concepts of efficiency, reliability and accuracy with examples.
- Find the solution of the initial value problem by using the Euler method.

$$y' = \frac{y-1}{2}, \quad y(0) = 1 \quad h = 0.1, 0.5$$

- Derive the Gregory Newton formula as a special case of Newton-divided difference formula.
- Prove that divided differences of order n of x^n are constant, where n is positive integer

Q2. a) Show that (i) $\Delta + I = E$, (ii) $(I - \nabla)^{-1} = E$

b) Apply Newton Forward difference formula to find missing value in the following:

(10)

x	0	1	2	3	4	5 $f(1)$
f(x)	3	8	15	?	47	47
		3	7	15	47	?

Q3. a) Find the Eigen value and Eigen vectors of

$$\begin{bmatrix} 2 & -2 & 3 \\ 2 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

b) Solve the initial value problem $y_{k+2} - 3y_{k+1} + 2y_k = 6k - 17$, $y_1 = 13$, $y_2 = 30$

(10)

Q4. a) Let the values as given in the following table. Find a polynomial which interpolate the given data

x:	-3	-2	-1	0	1
y:	16	7	4	1	-8

b) Compare the result of Trapezoidal and Simpson's $\frac{1}{3}$ rule for $\int_0^1 \frac{dx}{1+x^4}$, state which rule is better and why?

$$T = f(x) = \frac{h}{2} [y_0 + y_n + 2[y_1 + y_2 + y_3 + \dots + y_{n-1}]] \quad (10)$$

$$S = f(x) = \frac{h}{3} [y_0 + y_n + 3[y_1 + y_3 + y_5 + y_7 + \dots + y_{n-1}]] \quad (10)$$

Q5. a) Compute the first four derivative from the following data in the give table

x	2	5	9	13	15	21
f(x)	108243219	121550625	141158164	163047364	174900628	214358884

b) Solve the system of equations by Jacobi's method correct to four decimal places, using initial guess $x^{(0)} = y^{(0)} = z^{(0)} = 1$, $3x - y + z = 1$, $3x + 6y + 2z = 0$ and $3x + 3y + 7z = 4$.

Q6. a) Fit a polynomials corresponding to the following data using Lagrange interpolations formula

x	3	5	7	9	11
f(x)	6	24	58	108	174

$$f(x) = L_0 f(x) + L_1 f(x) + L_2 f(x) + \dots + L_n f(x)$$

b) Apply any iterative method to find approximate root of $x + e^{-x} \cos x = 0$.

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8th Spring Semester Examination 2025
Professional Practices (BS Computer Science)

below
below

Time Allowed: 2:30hrs

Max. Marks: 60

Note: Questions 1 is compulsory and attempt any Four questions from remaining five questions.

Q1. Write short note/Definition of the following: (20)

- i) Differentiate b/w profession and professionalism
- ii) Risks in software development projects
- iii) Security threats in Information Technology
- iv) Economic aspects of software development.

Q2. Define Professional Ethics and explain the importance of a Code of Ethics in computing professions. (10)

Q3. Define the competency. Discuss the benefits of long-life learning with example. (10)

Q4. What is a software contract? Describe its key components in detail. (10)

Q5. Discuss the structure and ethics of software house organization in detail. (10)

Q6. Describe the different types of Computer and Software Misuse. What are the ethical and legal implications of software piracy? (10)